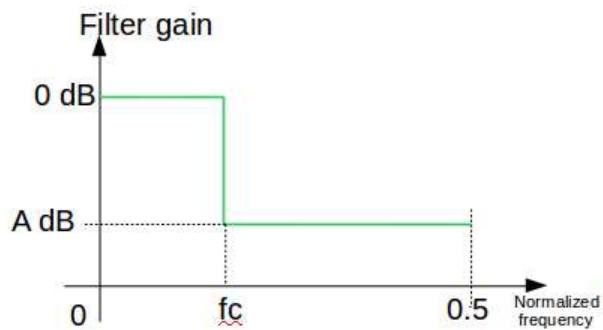


Commencé le mardi 28 mai 2024, 14:04**État** Terminé**Terminé le** mardi 28 mai 2024, 15:07**Temps mis** 1 heure 2 min**Points** 13,17/25,00**Note** 10,53 sur 20,00 (52,67%)**Question 1**

Terminé

Note de 0,00 sur 1,00

Consider a white noise entering the following filter:



For the filter, $A = -130$ dB and $f_c = 0,06$.

The white noise input power is 100 milliWatts.

What is the mean power of the noise after the filtering in milliWatt?

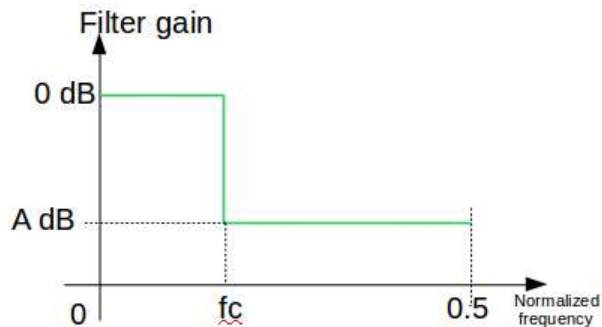
Réponse :

Question 2

Terminé

Note de 1,00 sur 1,00

Consider a sine wave entering the following filter:



For the filter, $A = -12$ dB and $f_c = 0.2$.

The sinus normalized frequency is 0,3 . its power is 100 milliWatts.

What is the mean power of the signal after the filtering in milliWatts?

Réponse :

Question 3

Terminé

Note de 0,00 sur 1,00

Consider the following signal $x(n) = s(n) + i(n)$ where $s(n)$ is the Signal of Interest (SOI) and $i(n)$ an interference.

The Signal to Interference ratio is -4,3 dB. The SOI power is 3,3 Watts.

The signal $x(n)$ will be filtered to reduce the effect of the interference $i(n)$. After filtering, the SOI, $s(n)$, must be untouched and the Signal to Interference ratio should be 15,3 dB.

What should be the expected attenuation of the filter in dB?

Réponse :

Question 4

Terminé

Note de 0,00 sur 1,00

in Python, consider the following code :

```
import numpy as np
n=np.arange(1024)
s=np.sin(2*np.pi*n*0.1)
b=np.random.randn(1024)
x=3,5*s+2,3*b
```

Knowing that $\text{np.cov}(b)=1$, what is the signal to noise ratio for the signal x in dB?

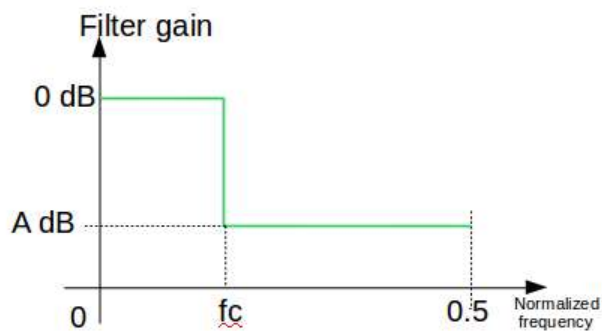
Réponse :

Question 5

Terminé

Note de 0,75 sur 1,00

Consider 2 sine waves entering the following filter:



For the filter, $A=-45$ dB and $f_c=0.2$.

The first sinus normalized frequency is 0,07 and the second is 0,43. The input power ratio between the first and the second is 10 dB.

What is their output power ratio in dB?

Réponse :

Question 6

Terminé

Note de 1,00 sur 1,00

Which one(s) of these statements are true ?

Veuillez choisir au moins une réponse.

- ☐ a. a IIR filter must have the module of its poles (roots of the $H(z)$ denominator) less than 1
- ☐ b. $y(n)=x(n)-x(n-1)-x(n-2)$ is a IIR filter
- ☐ c. $y(n)=x(n)-x(n-1)-y(n-1)$ is a IIR filter
- ☐ d. the IIR coefficients are equal to the impulse response
- ☐ e. A IIR filter is always stable

Question 7

Terminé

Note de 1,00 sur 1,00

Which one(s) of these statements are true ?

Veuillez choisir au moins une réponse.

- ☐ a. $y(n)=x(n)-y(n-1)$ is a FIR filter
- ☐ b. a FIR filter must have the module of its zeros (roots of the $H(z)$ numerator) less than 1
- ☐ c. $y(n)=x(n)-x(n-2)$ is a FIR filter
- ☐ d. A FIR filter is always stable
- ☐ e. the FIR coefficients are equal to the impulse response

Question 8

Non répondue

Noté sur 1,00

consider the following filter defined by its Z transfer function $H(z)=[6,8.+2,5.z-1]/[1-7,2.z-1]$

$h(n)$ is the impulse response of this filter. Give the value of $h(3)$.

Réponse :

Question 9

Terminé

Note de 1,00 sur 1,00

Considering the noise due to the quantization error, which one(s) of the following statements are true ?

Veuillez choisir au moins une réponse.

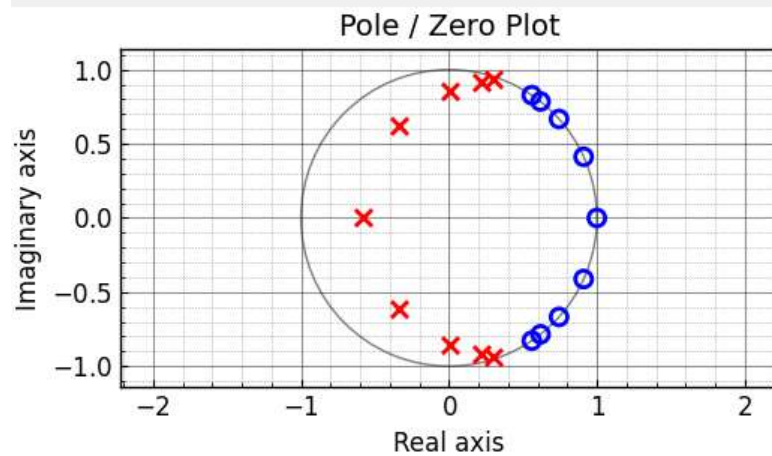
- ☐ a. the quantization errors are uncorrelated to each other
- ☐ b. The quantization error signal contains specific frequencies multiple of the sampling frequency
- ☐ c. the signal to noise ratio (SNR) can be approximated as $6.02 \times N$ where N is the number of bits. The unit is the Watt.
- ☐ d. The spectrum is Gaussian.
- ☐ e. The amplitude probability is uniform

Question 10

Terminé

Note de 0,42 sur 1,00

Consider a filter with the following zeros (o) and poles (x) :



What type of filter is it ?

Veuillez choisir au moins une réponse.

- ☐ a. It is a high filter
- ☐ b. it is a IIR filter
- ☐ c. it is a band pass filter
- ☐ d. it is stable filter.
- ☐ e. it is a low pass filter
- ☐ f. It is an unstable filter
- ☐ g. it is a band stop filter
- ☐ h. It is a FIR filter

Question 11

Terminé

Note de 0,00 sur 1,00

Consider an ADC with an input dynamic range between 0V and 10 V.

The ADC output is coded over 12 bits as unsigned integer.

What will be the digital value provided by the ADC for an analog input of 9,4 V ?

Réponse :

Question 12

Terminé

Note de 0,00 sur 1,00

Consider the following filter : $y(n) = x(n) + x(n-2) + x(n-3) - y(n-2) + y(n-3)$

what is the correct Z transfert fonction, $H(z)$, of this filter ?

Veuillez choisir au moins une réponse.

- ☐ a. $H(z) = \frac{1-z^{-2}+z^{-3}}{1+z^{-2}+z^{-3}}$
- ☐ b. $H(z) = \frac{1+z^{-2}+z^{-3}}{1-z^{-2}+z^{-3}}$
- ☐ c. $H(z) = \frac{1-z^{-2}-z^{-3}}{1+z^{-2}-z^{-3}}$
- ☐ d. $H(z) = \frac{1+z^{-2}-z^{-3}}{1+z^{-2}+z^{-3}}$
- ☐ e. $H(z) = \frac{1+z^{-2}+z^{-3}}{1+z^{-2}-z^{-3}}$

Question 13

Terminé

Note de 1,00 sur 1,00

Consider the following filter : $y(n) = x(n) - 2x(n-1) + x(n-2) - 0,8y(n-2)$

Type 0 if the filter is stable, 1 if not.

Réponse :

Question 14

Terminé

Note de 1,00 sur 1,00

In Python, the filter coefficients are stored into 2 arrays :

- array b for the non recursive coefficients
- array a for the recursive coefficients

Among the following proposition, which one(s) will provide the frequency response $H(f)$?

np is the alias for the numpy library and sci is the alias for the scipy.signal library.

Veuillez choisir au moins une réponse.

- ☐ a. `n=np.arange(1024)`
`x=np.sin(2*np.pi*n*0.1)`
`y=sci.lfilter(b,a,x)`
`f,H=psd(y,NFFT=256,Fs=1)`
- ☐ b. `x=np.random.randn(512)`
`y=sci.lfilter(b,a,x)`
`f,H=psd(y,NFFT=256,Fs=1)`
- ☐ c. `f,H=sci.freqz(a,b,NWord=512,Fs=1)`
- ☐ d. `x=np.zeros(512)`
`x[0]=1`
`y=sci.lfilter(b,a,x)`
`H=np.fft.fft(y)`

Question 15

Terminé

Note de 1,00 sur 1,00

Consider the following 4-bits signed binary number 1111 (two's complement coding). Extend to 8 bits this number so that it keeps the same decimal value.

Réponse :

Question 16

Terminé

Note de 1,00 sur 1,00

Consider the following 4-bits signed binary number 0010 (two's complement coding). Extend to 8 bits this number so that it keeps the same decimal value.

Réponse : **Question 17**

Terminé

Note de 1,00 sur 1,00

Consider the following binary number 1000. What is its decimal signed value assuming that we use the two's complement coding?

Réponse : **Question 18**

Terminé

Note de 1,00 sur 1,00

Consider the following binary number 0100. What is its decimal signed value assuming that we use the two's complement coding?

Réponse : **Question 19**

Terminé

Note de 1,00 sur 1,00

Consider the following binary number 1000. What is its decimal unsigned value?

Réponse :

Question 20

Terminé

Note de 1,00 sur 1,00

Consider the following binary number 0110. What is its decimal unsigned value?

Réponse : **Question 21**

Terminé

Note de 0,00 sur 1,00

Consider the following signed fixed point representation number 10.11 : What is its decimal signed value assuming that we use the two's complement coding?

Réponse : **Question 22**

Terminé

Note de 0,00 sur 1,00

code the following decimal number 2.74591874 in signed fixed point representation over 16 bits :

Réponse : **Question 23**

Terminé

Note de 0,00 sur 1,00

code the following decimal number -2.74591874 in signed fixed point representation over 16 bits :

Réponse :

Question **24**

Terminé

Note de 0,00 sur 1,00

What is the binary representation of $\exp(1)=2.718281828459045\dots$ in $Q(16,10)$ format ?

use a dot to separate the integer part from the fractional part

Réponse :

Question **25**

Terminé

Note de 0,00 sur 1,00

What is the binary representation of $\exp(1)=2.718281828459045\dots$ in $Q(16,13)$ format ?

use a dot to separate the integer part from the fractional part

Réponse :